# TRAVEL PLANNER

# PYTHON MINI PROJECT

**A simple desktop application for trip planning, itinerary management and budget estimation**

| DESTINATIONS | ITINERARY | BUDGET | TRIP SUMMARY |
|---|---|---|---|
| Explore places | Plan activities | Estimate costs | Save your plan |

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# PROJECT OVERVIEW & OBJECTIVES

## PROJECT OVERVIEW

A Python desktop application that helps users organize a trip by entering a destination, travel dates, activities and estimated expenses.

## PROBLEM ADDRESSED

Trip details are often scattered across notes and messages. Planning them in one place makes the schedule and estimated budget easier to manage.

## PROJECT OBJECTIVES

- Record destination and travel dates
- Organize activities day by day
- Estimate transport, stay and food costs
- Save and retrieve trip details

## EXPECTED OUTCOME

- Clear itinerary and trip summary
- Quick budget estimate
- Reusable saved trip information
- Simple, user-friendly interface

# TECHNOLOGY STACK

### PYTHON

Core language for trip logic, calculations and data handling.

### TKINTER

Creates the desktop GUI with forms, buttons and display areas.

### DICTIONARIES

Store destination details, itinerary entries and expense categories.

### FUNCTIONS

Separate tasks such as adding trips, calculating costs and saving plans.

### FILE HANDLING

Save and load trip information from a local text or JSON file.

### EXCEPTION HANDLING

Handles invalid input and file errors with helpful messages.

**Suggested libraries: tkinter, json, datetime (built-in Python modules)**

# MAIN FEATURES

### Create Trip

Enter trip name, destination and travel dates.

### Plan Itinerary

Add places to visit and activities for each day.

### Manage Expenses

Enter estimated travel, hotel, food and activity costs.

### Calculate Budget

Automatically calculate the estimated total.

### Save Trip

Store trip details in a local file for later use.

### View Summary

Display destination, dates, itinerary and total budget.

# SYSTEM WORKFLOW

1

## Launch App

Python opens the Tkinter window.

2

## Enter Details

User enters destination, dates and trip name.

3

## Add Activities

Places and activities are added to the itinerary.

4

## Add Expenses

Estimated costs are entered by category.

5

## Calculate Budget

The program totals the expenses.

6

## Save & Review

Trip data is saved and the summary is displayed.

Validation checks required fields, date format and non-negative expense values.

# SAMPLE TRIP PLAN & BUDGET

| NO. | DESTINATION / CATEGORY | PLAN / ACTIVITY | ESTIMATED COST |
|---|---|---|---|
| 1 | Ahmedabad | Heritage walk | ₹800 |
| 2 | Udaipur | City Palace & Lake Pichola | ₹2,500 |
| 3 | Mount Abu | Nakki Lake & viewpoints | ₹1,800 |
| 4 | Travel | Transport estimate | ₹1,500 |
| 5 | Accommodation | 2-night stay estimate | ₹3,000 |
| 6 | Food & misc. | Meals and local expenses | ₹1,200 |

Illustrative example only — costs are sample estimates and can be edited by the user.

**Estimated total: ₹10,800**

# DATA DESIGN

### TRIP DETAILS

- trip_id
- trip_name
- destination
- start_date / end_date

### ITINERARY

- day / date
- place_name
- activity
- notes

### EXPENSES

- category
- description
- estimated_cost
- total_budget

**Relationship: one trip can contain multiple itinerary activities and expense entries.**

Data can be represented using nested Python dictionaries and lists, then saved to a JSON or text file.

# PYTHON PROGRAM STRUCTURE

### TRAVEL PLANNER FUNCTIONS

Functions handle creating trips, adding itinerary items, entering expenses, calculating totals and displaying summaries.

### TKINTER GUI

The interface collects input through labels, entry fields, buttons and list or text widgets.

### DICTIONARIES

Store trip, activity and expense details.

### FILE HANDLING

Use JSON/text files to save and reload trips.

### VALIDATION

Check missing fields, dates and expense values.

Entry point: main() creates the application window and starts the Tkinter event loop.

# SAMPLE OUTPUT — TRIP SUMMARY

## TRAVEL PLANNER — TRIP SUMMARY

Trip: Weekend Getaway  Destination: Udaipur  Duration: 3 days

DAY 1 • City Palace and old city
DAY 2 • Lake Pichola and local markets
DAY 3 • Sajjangarh viewpoint and return

Transport: ₹1,500
Stay: ₹3,000
Food: ₹1,200
Activities: ₹2,500

**ESTIMATED TOTAL: ₹8,200**

Sample output for demonstration; destination and costs are editable.

# CONCLUSION & FUTURE SCOPE

## CONCLUSION

The Travel Planner demonstrates Python GUI development, dictionaries, functions and file handling. It brings trip details, activities and budget estimates into one simple application.

## FUTURE ENHANCEMENTS

- Weather and map integration
- Hotel and transport booking links
- Currency conversion
- Export itinerary to PDF
- User accounts and cloud sync

## KEY LEARNING

A practical demonstration of GUI design, structured programming, data organization, file persistence and input validation in Python.

## THANK YOU